

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_direct_generate_plus_one_repair.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.04336	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	76.18308	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1380.50626	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	1.0	0.070487	0.070487	316.13256	2	0	0	0.849461	15.755
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	1.0	0.070832	0.070832	348.21898	2	0	0	0.932473	13.97
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.071239	0.071239	246.68874	2	0	0	0.798138	13.245

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.04336.
- Accepted epoch count: 0.

- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 76.18308.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1380.50626.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.

- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.068916.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070487.
- Held-out runtime ms: 316.13256.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.004296.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.099076, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.094328, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.056955, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.056596, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.045478, errors=[]

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.068916.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070832.
- Held-out runtime ms: 348.21898.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.004511.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.099076, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.095536, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.056955, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.056596, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.045997, errors=[]

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.067412.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.071239.
- Held-out runtime ms: 246.68874.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.003638.
- Worst held-out instances:
 - X-n176-k26: feasible=True, penalized gap=0.098323, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.095053, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061779, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.056154, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.044885, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary

Condition	Feasibility Rate	Heldout Feasible Gap	Runtime (ms)	Robustness Dispersion	No velty	Complexity	Notes
Baseline Nearest Neighbor Constructive	1.0	0.2734	42.0	0.0028	0.0	0.0	Fastest, largest gap
Baseline Clarke-Wright Savings	1.0	0.0738	76.2	0.0078	0.0	0.0	Strong baseline, moderate runtime
Baseline Regret Insertion Local Search	1.0	0.4664	1380.5	0.0869	0.0	0.0	Worst gap, longest runtime, least robust
Direct Generate + One Repair (Learned)	1.0	**0.0705**	316.1	0.0043	0.85	15.8	Best gap overall, moderate runtime
Budget-Matched No Replay (Learned)	1.0	0.0708	348.2	0.0045	0.93	14.0	Similar gap to direct + repair
Replay Solver Evolution (Learned with Replay)	1.0	0.0712	246.7	0.0036	0.80	13.2	Slightly worse gap than no replay, faster runtime, most robust

Key Points

- **Feasibility:** All conditions achieved perfect (100%) feasibility on held-out data.
- **Objective Gap:**
 - Learned methods outperform fixed baselines Clarke-Wright and Nearest Neighbor in gap, particularly direct generate plus one repair (0.0705 gap) slightly beating Clarke-Wright (0.0738 gap).
 - Regret insertion local search baseline performs worst in gap (0.4664).
- **Runtime:**
 - Nearest neighbor is fastest (42ms), but with high gap.
 - Replay solver is fastest among learned methods (~247ms), direct+repair and no replay require 316-348ms.
 - Regret insertion is slowest (~1380ms).
- **Robustness:**
 - Replay solver has lowest robustness dispersion (0.0036), indicating consistent performance across structure families.
 - Learned methods show low dispersion, better than most baselines except nearest neighbor.
- **Learned vs Fixed Baselines:**
 - Direct generate plus one repair (learned) clearly beats fixed baselines in objective gap and robustness with a moderate runtime cost.
- **Replay vs No Replay (Both Budget-Matched):**
 - Replay solver has a slightly worse feasible gap (~0.0712 vs ~0.0708) but better runtime (247ms vs 348ms) and better robustness dispersion (0.0036 vs 0.0045).
 - No clear dominant advantage in gap for replay, but replay improves robustness and runtime.

Conservative Conclusion

The **direct generate plus one repair** learned condition produces the best overall solution quality (lowest heldout gap), outperforming fixed baselines Clarke-Wright and nearest neighbor with full feasibility and reasonable runtime. Replay-aware iterative search (replay solver evolution) does not clearly improve objective gap over budget-matched no-replay learned search but provides better runtime efficiency and robustness, suggesting replay helps stabilize and accelerate search though its superiority in solution quality is not demonstrated.